DayOS

Daily Operating System for the Focused Researcher

Product Requirements Document — v1.1

Version	1.1 — Iterated from Open Questions
Platform	Progressive Web App (PWA)
Audience	MTech Robotics — IISc Bengaluru
Primary Device	Mobile First (iOS & Android)
Backend	Supabase (PostgreSQL + Auth + Realtime)
Status	Draft — Pending Developer Handoff

1. Executive Summary

DayOS is a mobile-first Progressive Web App designed specifically for graduate researchers juggling academic workloads, fitness goals, and deep research. The product replaces the fragmented habit of using separate apps for gym tracking, food logging, study timers, and project management with a single, opinionated daily command centre.

The app is task-checklist driven — meaning the primary interaction model is a structured daily card that the user works through each day, not a passive analytics dashboard. Everything is logged in context, inline, and fast. End-of-day data is synced to Supabase for persistence and cross-device access, with full offline-first support via a service worker.

2. Goals & Non-Goals

2.1 Goals

- Give the user a single, zero-friction daily view of everything that matters: workouts, nutrition, classes, study blocks, research tasks, and journalling.
- Ensure the app works completely offline — all data is written locally first, then synced to Supabase in the background.
- Be installable as a home-screen PWA on both Android and iOS with no App Store required.
- Keep logging fast — any single log action should require no more than 3 taps or inputs.
- Surface lightweight progress signals (streaks, weekly summaries) without overwhelming the user with charts.

2.2 Non-Goals (v1)

- Social or sharing features.
- AI-generated recommendations or coaching.
- Integration with wearables (Garmin, Apple Watch) — deferred to v2.
- Custom recurring reminders per-habit — v1 supports a single configurable daily reminder time.
- Multi-user or team features.

3. User Persona

The IISc Researcher-Athlete

Age: 22–26 · Degree: M.Tech Robotics · Location: IISc Campus, Bengaluru

Has 3–5 hours of structured coursework per day, plus research lab time that is largely self-directed. Trains 4–5 days a week. Eats primarily at the mess with occasional outside food. Reads 2–4 research papers per week. Struggles to maintain consistency on all fronts simultaneously — a good gym week often means a bad study week and vice versa.

Core frustration: Too many apps, too much friction. Needs one place to see if today is ‘on track’ or not.

4. Information Architecture

The app has five top-level tabs accessible via a persistent bottom navigation bar:

Tab	Icon	Description
Today	🏠 Home	The daily checklist — the app's primary view. One scrollable card per module.
Schedule	📅 Calendar	Timetable view of the week with classes, lab slots, and deadlines.
Research	🔬 Flask	Project tracker with tasks, paper log, and lab hours.
Stats	📊 Chart	Weekly summaries for each module. Streaks and completion rates.
Settings	⚙️ Gear	Profile, Supabase sync status, notification preferences, data export.

5. Module Specifications

5.1 Today — The Daily Checklist

The Today screen is the app's centrepiece. It renders a vertically scrollable list of module cards for the current day. Each card has a header with a completion ring and an expand/collapse chevron. Cards are ordered by the user's configured priority.

Behaviour

- Cards default to expanded on first open of the day, and remember their collapse state after.
- A 'Day complete!' celebration state is shown when all mandatory items in all cards are checked off.
- A persistent top banner shows the date, day streak, and a motivational one-liner (rotated daily from a static list).
- Pull-to-refresh triggers a Supabase sync.

5.2 Workout & Fitness Card

Features

- User pre-configures a weekly training split (e.g., Push/Pull/Legs/Rest). The card shows today's session type automatically.
- Exercises are added as a checklist. Each item shows: Exercise name, sets × reps, weight (optional).
- Long-press on a set to log actual values (vs planned), enabling progressive overload tracking.
- A 'Rest day' card variant replaces the checklist with a recovery prompt and optional note.
- Weekly volume summary available in Stats tab.

Data Fields

Field	Type	Notes
session_type	enum	push / pull / legs / upper / lower / cardio / rest
exercises	array	Each: { name, planned_sets, planned_reps, weight_kg, logged_sets[] }
duration_mins	integer	Total session duration
notes	text	Optional free-text
completed	boolean	Marks the workout as done for the day

5.3 Macro & Nutrition Card

Features

- User sets daily macro targets: calories, protein (g), fats (g), carbs (g). These are stored in user_preferences.
- Meals can be logged in two ways: (1) Manual entry — type a meal name and enter values for each macro column directly. (2) Markdown table import — user pastes a markdown-formatted table; the app parses it and creates one meal entry per row automatically.
- Calories in the import table are used as-is. Auto-computation from macros (P×4 + C×4 + F×9) is a fallback only, used when the Calories column is empty or absent.
- Frequently used meals are saved as templates for 1-tap reuse.
- A horizontal macro progress bar (Protein · Fats · Carbs) with calorie total updates live as meals are added.
- Water intake tracker: quick +200ml / +500ml buttons with a daily goal ring.

Markdown Import Specification

The import parser must handle tables in the following exact format, which the user will paste from their own nutrition log:

```
Expected column order (case-insensitive header match):
Item | Portion | Calories | Protein (g) | Fats (g) | Carbs (g)
Example rows:
Naan (4 small halves) | 2 whole | 380 | 10 | 10 | 62
Shahi Paneer Cubes | ~5-6 pieces | 100 | 8 | 7 | 2
TOTAL |  | 1,410 | 51g | 61g | 172g
```

Parser Rule	Detail
Column delimiter	Pipe character | with flexible surrounding whitespace
Header row	First row; matched case-insensitively. 'Protein (g)', 'Fats (g)', 'Carbs (g)'

Parser Rule	Detail
	are the canonical names.
Separator row	Row of dashes (---|---|---) is detected and skipped automatically
TOTAL row	Any row where Item column equals 'TOTAL' (case-insensitive) is skipped — not logged as a meal entry
Portion column	Stored as a plain string. Not parsed or computed.
Numeric cleaning	Commas stripped (1,410 → 1410). Unit suffixes stripped (51g → 51). Tilde prefix stripped (~5 → 5 but stored as label).
Calories fallback	If Calories cell is empty or 0, compute as: (Protein × 4) + (Carbs × 4) + (Fats × 9)
Validation	Warn (non-blocking) if a row has fewer than 4 filled numeric columns. Skip silently if Item name is empty.
Import UX	After paste, show a preview table of parsed rows before confirming. User can deselect individual rows before saving.

Data Fields

Field	Type	Notes
meals	array	Each: { name, portion_label, protein_g, fats_g, carbs_g, calories, source: manual|import }
water_ml	integer	Cumulative for the day
targets	object	{ calories, protein_g, carbs_g, fat_g } — user profile
meal_templates	array	Saved quick-log templates

5.4 Study Sessions & Pomodoro Card

Features

- User creates study blocks for the day: subject/course name, target duration, and optional topic.
- Each block has an embedded Pomodoro timer (default: 25 min work / 5 min break, configurable).
- The Pomodoro timer pauses automatically when the app is backgrounded on mobile. It does not continue silently. A local notification fires when the timer is paused due to backgrounding, prompting the user to return.
- Completed pomodoros are shown as filled tomato icons on the block.
- Total focused time for the day is shown in the card header.
- Subjects are colour-tagged and carry over to the Schedule tab for consistency.

Data Fields

Field	Type	Notes
blocks	array	Each: { subject, topic, target_mins, pomodoros_done, completed }
pomodoro_config	object	{ work_mins, break_mins, long_break_mins, cycles_before_long }
total_focus_mins	integer	Computed: sum of actual time across blocks

5.5 Timetable & Class Schedule

Features

- User inputs their semester timetable once (course name, day of week, start/end time, room/mode).
- The Today card shows today's classes as a read-only timeline.
- The Schedule tab shows a full weekly grid view.
- One-time events and deadlines (exams, assignment submissions, lab reports) can be added to any date.
- Deadlines within the next 7 days surface as a pinned banner on the Today screen.

Data Fields

Field	Type	Notes
recurring_classes	array	Each: { course, day_of_week, start_time, end_time, room, mode }
events	array	Each: { title, date, time, type: deadline|exam|other, notes }

5.6 Research & Project Progress

Features

- User creates one or more Projects (e.g., 'M.Tech Thesis', 'Course Project — ME256').
- Each project has a Kanban-lite task list with three columns: To Do, In Progress, Done.
- Today's research card shows tasks marked 'In Progress' as the active checklist for the day.
- A Paper Reading Log is attached to each project: title, arXiv ID, status (to-read / reading / done), date read, and a notes field.
- arXiv ID autofill: when the user enters an arXiv ID (e.g., 2304.01234), the app calls the arXiv API to auto-populate the paper title, authors, and abstract. Works online only; degrades gracefully to manual entry when offline.
- Task completion rates are shown in the Stats tab per project.

- Note: lab hours are not tracked. Research tasks and paper reading are the sole units of research progress in DayOS.

Data Fields

Field	Type	Notes
projects	array	Each: { name, description, colour_tag, tasks[], papers[] }
tasks	array	Each: { title, status, due_date, notes, project_id }
papers	array	Each: { title, authors, abstract, arxiv_id, status, date_added, date_read, notes }

5.7 Daily Journal & Reflection

Features

- A single free-text journal card at the bottom of Today's checklist.
- Three optional prompt fields to structure the entry: 'Highlight of the day', 'What slowed me down', and 'Tomorrow's #1 priority'.
- User can toggle between prompted mode and a blank free-write mode.
- Entries are searchable from the Stats tab by keyword.

Sunday Planning Mode

- Every Sunday, the journal card expands into a full Sunday Review & Weekly Plan.
- The review section contains: 'Weekly highlight', 'What I'd do differently', and a read-only summary of the week's completed tasks across all modules.
- The planning section lets the user set intentions for the coming week across each module: 3 research tasks to move forward, study subjects to focus on, workout split for the week, and one personal goal.
- Weekly plan data is stored and surfaced as a collapsible reference banner on each day of the following week's Today screen.

Data Fields

Field	Type	Notes
date	date	ISO date for the entry
highlight	text	Optional prompted field
blockers	text	Optional prompted field
top_priority_tomorrow	text	Optional prompted field
free_text	text	Unstructured journal body

Field	Type	Notes
weekly_review	text	Sunday only: reflection field
weekly_plan	JSONB	Sunday only: { research_tasks[], study_subjects[], workout_split, personal_goal }

5.8 Quick Scratchpad

Features

- A floating action button (FAB) visible on all screens, positioned above the bottom nav bar.
- Tapping it opens a lightweight bottom sheet modal with a single text area. No titles, no categories — just write.
- Each note is saved with a timestamp and stored in a dedicated scratchpad log.
- The full scratchpad history is accessible from the Stats tab under a 'Notes' section, searchable by keyword.
- Notes are local-first (IndexedDB) and sync to Supabase in the background like all other data.

Data Fields

Field	Type	Notes
content	text	The note body
created_at	timestamp	Auto-set on creation
context_screen	text	Which tab was open when note was created (optional, for context)

6. PWA & Technical Requirements

6.1 Offline-First Architecture

- All writes go to IndexedDB first. A background sync queue (via Service Worker) flushes to Supabase when connectivity is restored.
- The app must be fully functional with no network connection. The only degraded state is sync lag.
- Conflict resolution: last-write-wins per record, keyed on updated_at timestamp.

6.2 Installability

- Must include a valid Web App Manifest: name, short_name, icons (192px, 512px, maskable), theme_color, background_color, display: standalone, orientation: portrait.
- Lighthouse PWA score target: ≥90.
- App should prompt install on second session or after one week of use (not on first launch).

6.3 Push Notifications

- Single configurable daily reminder: user sets a time (e.g., 7:00 AM). Notification reads: 'Your DayOS is ready — tap to start your day.'
- Optional workout reminder if no workout logged by a user-set cutoff time (e.g., 7:00 PM).
- Optional Pomodoro session end alerts (local notifications only, no server push needed for this).
- Notifications use the Web Push API with a Supabase Edge Function as the VAPID push server.

6.4 Backend — Supabase

A single personal Supabase project is sufficient for v1. No multi-environment (dev/prod) setup is required. The developer should use a single project with RLS enforced throughout.

Service	Usage
Auth	Magic link email auth (no password). Single user assumed for v1.
PostgreSQL	Primary persistence layer. All module data tables.
Realtime	Optional — for future multi-device live sync.
Edge Functions	VAPID push notification delivery.
Storage	Not required in v1.

7. UX & Design Guidelines

7.1 Visual Language

Token	Value
Primary	#3A86FF — Electric Blue
Background	#FFFFFF — Pure White
Surface	#F7F8FA — Off-White (cards)

Token	Value
Border	#E8E8E8 — Hairline
Text Primary	#1A1A2E — Near Black
Text Secondary	#6B7280 — Muted Gray
Success	#10B981 — Emerald
Warning	#F59E0B — Amber
Font	Inter (Google Fonts) — 400, 500, 600, 700
Border Radius	12px for cards, 8px for inputs, 999px for pills/buttons

7.2 Mobile UX Constraints

- Bottom navigation bar — no hamburger menus.
- All tap targets ≥48px height.
- Cards use full-width layout with 16px horizontal padding.
- No horizontal scrolling on the Today screen.
- Swipe-to-complete on checklist items (right swipe = done, left swipe = delete).
- Haptic feedback on task completion (if supported by device).

8. Supabase Schema Overview

All tables include id (UUID), user_id (FK to auth.users), created_at, and updated_at. Row Level Security (RLS) must be enabled on all tables, restricting access to the owning user_id.

Table	Key Columns
daily_checklist	date, status (complete/partial/missed), workout_done, nutrition_logged, study_done, journal_done
workouts	date, session_type, exercises (JSONB), duration_mins, notes
nutrition_logs	date, meals (JSONB), water_ml, import_source (manual|markdown)
meal_templates	name, protein_g, fats_g, carbs_g, calories (auto-computed fallback)
study_blocks	date, subject, topic, target_mins, pomodoros_done, paused_at
timetable	course, day_of_week, start_time, end_time, room, mode
events	title, date, time, type (deadline/exam/other), notes
projects	name, description, colour_tag
tasks	project_id, title, status, due_date, notes
papers	project_id, title, authors, arxiv_id, status, date_read, notes

Table	Key Columns
journal_entries	date, highlight, blockers, top_priority_tomorrow, free_text, weekly_reflection
scratch_notes	id, content, created_at, pinned
sunday_plans	week_start_date, workout_intentions, study_intentions, research_intentions, weekly_goal
exam_mode_config	active, exam_title, exam_date, hidden_modules (JSONB)
user_preferences	daily_reminder_time, workout_cutoff_time, pomodoro_config (JSONB), macro_targets (JSONB), card_order (JSONB)

9. Recommended Tech Stack

Layer	Choice	Rationale
Frontend	React + Vite	Fast build, great PWA plugin ecosystem
Styling	Tailwind CSS	Utility-first, easy mobile-first layout
State	Zustand + React Query	Lightweight global state + async cache
Offline DB	Dexie.js (IndexedDB)	Simple IndexedDB wrapper with sync hooks
Service Worker	Workbox (via vite-plugin-pwa)	Precaching, background sync, push
Backend	Supabase JS SDK v2	Auth, DB, Realtime, Edge Functions
Push	Web Push + Supabase Edge Fn	VAPID keys, server-side delivery
Hosting	Vercel or Netlify	Easy PWA deploy, HTTPS by default

10. Suggested Development Milestones

Sprint	Focus	Deliverable
1	Foundation	Vite PWA scaffold, Supabase auth (magic link), IndexedDB + Dexie setup, bottom nav shell
2	Today Screen	Daily checklist card container, missed-day marking logic, timetable display, deadline banner
3	Fitness & Nutrition	Workout card, macro card with MD table import parser, progress bars, meal templates, water tracker
4	Study & Research	Pomodoro card (with backgrounding pause), study blocks, project/task manager, paper log + arXiv autofill

Sprint	Focus	Deliverable
5	Journal & New Modules	Journal card, Sunday planning mode, Quick Scratchpad FAB, Exam Mode toggle
6	Stats & Polish	Stats tab (weekly summaries, streaks, missed-day heatmap), Sunday review surfacing
7	PWA & Push	Service worker, offline sync, push notifications, install prompt, Lighthouse audit ≥90
8	Beta & Bugfix	Device testing (Android + iOS Safari), performance tuning, Settings screen, edge cases

11. New Modules (Added in v1.1)

11.1 Quick Scratchpad

A persistent floating action button (FAB) visible on every screen. Tapping it opens a bottom-sheet note editor for zero-friction capture of ideas, paper references, or task reminders mid-class — without losing your place in the app.

Features

- FAB sits just above the bottom nav bar. Tapping opens a bottom-sheet modal with a single text area.
- Notes save automatically as the user types (debounced 500ms). No save button.
- All scratch notes are listed in a ‘Notes’ sheet accessible from the FAB long-press.
- Notes can be pinned to stay at the top, or promoted into a Research task or Journal entry via a long-press context menu.
- Fully offline-first. Syncs to Supabase with all other data.

Data Fields

Field	Type	Notes
id	UUID	Primary key
content	text	The note body
pinned	boolean	Whether pinned to top of list
created_at	timestamp	Auto-set on creation
promoted_to	enum | null	task / journal / null

11.2 Sunday Planning Mode

Every Sunday, a ‘Plan Your Week’ card surfaces at the top of the Today screen. A lightweight weekly intention-setting flow — not a rigid schedule builder. The resulting plan stays visible as a collapsed banner throughout the week.

Features

- Auto-triggered on Sundays. Also accessible manually from Settings at any time.
- Four short intention fields: (1) Workout plan — which sessions this week, (2) Study focus — subjects or topics to prioritise, (3) Research goal — one concrete output for the week, (4) Weekly intention — one free-text sentence about the week’s theme.
- Once completed, intentions display as a read-only collapsed banner on Today for Monday–Saturday.
- Past Sunday plans are archived in the Stats tab for weekly retrospectives.

Data Fields

Field	Type	Notes
week_start_date	date	ISO date of the Monday of that week
workout_intentions	text	E.g. ‘PPL + 1 cardio session’
study_intentions	text	E.g. ‘Finish ME256 module 4’
research_intentions	text	E.g. ‘Complete literature review draft’
weekly_goal	text	One-sentence theme for the week

11.3 Exam Mode

A toggle in Settings and via a chip on the Today banner. When active, Exam Mode strips the Today screen to study essentials — removing workout and nutrition noise during crunch periods. Data is never deleted; modules are only hidden from the Today view.

Behaviour

- When ON: Today screen shows only the Study card, Research card (tasks only), and Journal card. Workout and Nutrition cards are hidden from Today but remain accessible via their respective tabs.
- The Today banner shows a red ‘EXAM MODE’ chip with a live countdown: e.g. ‘Exam in 3 days’.
- User sets an exam title and date on activation. Multiple exams can be queued.
- Exam Mode auto-deactivates the day after the exam date, with a confirmation prompt before turning off.

Data Fields

Field	Type	Notes
active	boolean	Whether Exam Mode is currently on
exam_title	text	E.g. ‘ME256 Mid-sem’
exam_date	date	Auto-deactivation trigger date
hidden_modules	JSONB	Array of module keys e.g. [‘workout’, ‘nutrition’]

12. Changelog

Version	Date	Summary
v1.1	Current	Resolved all 6 open questions. Removed lab hours entirely. Added Quick Scratchpad, Sunday Planning Mode, Exam Mode (focus strip). Added markdown nutrition import + arXiv autofill. Updated schema, milestones, and decisions log.
v1.0	Prior	Initial PRD. 7 modules specced. Supabase + offline-first architecture defined. Tech stack selected.

13. Decisions Log

All open questions from PRD v1.0 have been resolved. The following decisions are locked for v1 and should not require further discussion with the developer.

Decision	Resolution
Pomodoro backgrounding	Timer pauses when app is backgrounded. A local notification fires to prompt the user to return. No background countdown.
Lab hours tracking	Removed entirely. Not relevant to the user’s workflow. Research progress is tracked via tasks and paper reading only.
Calorie & macro entry	Two modes: manual entry and markdown table paste. Calories in the table are used directly. Auto-compute (P×4 + C×4 + F×9) is fallback only when Calories column is empty. Column order: Item | Portion | Calories | Protein | Fats | Carbs.
Missed days	A day with no logged checklist activity is marked as ‘missed’. Missed days break the streak. They are not skipped or ignored in the Stats view.
Supabase environment	A single personal Supabase project is sufficient for v1. No dev/prod split required.
arXiv autofill	Supported. Entering an arXiv ID triggers an API call to auto-fill

Decision	Resolution
	title, authors, and abstract. Degrades gracefully to manual entry when offline.

✅ **PRD v1.1 — All decisions locked. Ready for developer handoff.**